Practice Set 9 Paper 2 – Urology

Contents

1. Q1. Diagnosis and Treatment of renal trauma (LAQ, 25)
2. Q2. Discuss etiopathogenesis of uric acid calculi. Describe rationale of medical management, (LAQ, 25)
3. Q3a. Methods for GFR estimation in prospective tx donor. (SAQ, 10)
4. Q3b. Measures to decrease complications of TURP (SAQ, 10)
5. Q3c. What is optical coherence tomography. What is its potential role in management of CA bladder? (SAQ, 10)
6. Q3d. Tubeless PCNL (SAQ, 10)
7. Q3e. Medullary sponge kidney (SAQ, 10)
8. Q3f. Amendments to THOA: 2011. (SAQ, 10)

Practice paper compiled from the question bank.

Q1. Diagnosis and Treatment of renal trauma

Type: LAQ (25 marks) (also answered in 2017-Summer Paper 2, Q2; grade IV in Practice Set 3 Paper 2, Q3c)

Introduction

- The kidney is the **commonest urological organ injured** (up to 5% of all trauma). **Blunt injury causes 80–90%**; firearms cause most penetrating injuries.
- **Rapid deceleration** stretches the kidney at its fixed points (hilum and UPJ, ureteropelvic junction): arterial intimal tear and thrombosis, vein disruption, pedicle or UPJ avulsion.

Diagnosis

History and examination

- **Mechanism is the key history**. Signs: flank bruise or tenderness, **fractured ribs 9–12**, flank or low chest wound.

Haematuria

- Test the **first urine sample**. Significant: gross haematuria, or microhaematuria (>5 RBC [red blood cells] per high-power field) with **shock (SBP [systolic blood pressure] <90 mmHg)**.
- Degree does **not** match severity. It may be **absent in pedicle injury** and in about half of UPJ injuries.

Indications for imaging (AUA [American Urological Association] and EAU [European Association of Urology])

1. Blunt trauma with gross haematuria, or microhaematuria with SBP <90 mmHg at any time.
2. Rapid deceleration, even without haematuria.
3. Penetrating trauma with likely renal injury, any haematuria.
4. Children: image liberally (hypotension appears late).

- Microhaematuria alone without shock: observe.

Imaging

- **Contrast-enhanced CT (computed tomography) is the gold standard.** The arterial or nephrographic phase (~80 s) shows bleeding; the **delayed excretory phase (5–10 min)** shows urinary extravasation.

CT finding	Suggests
Medial haematoma	Pedicle (vascular) injury
Medial extravasation, no contrast in distal ureter	Renal pelvis or UPJ avulsion
Absent enhancement (cortical rim sign)	Main renal artery occlusion
Active contrast blush; haematoma rim >4 cm	Bleeding likely to need intervention

- **One-shot IVP (intravenous pyelography)** on the table if unstable: 2 mL/kg, one film at 10 min; confirms a **functioning contralateral kidney**.
- **Also know:** FAST (focused assessment with sonography for trauma) is non-specific; CT shows renal vein injury poorly.

Classification: AAST (American Association for the Surgery of Trauma) 2018

Grade	Injury
I	Subcapsular haematoma and/or contusion, no laceration
II	Perirenal haematoma within Gerota's fascia; laceration ≤1 cm, no extravasation

Grade	Injury
III	Laceration >1 cm, no collecting-system injury; vascular injury or active bleeding **contained within Gerota's fascia**
IV	Laceration into collecting system with extravasation; renal pelvis laceration or UPJ disruption; **segmental artery or vein injury**; active bleeding beyond Gerota's fascia; segmental or complete infarction
V	**Main renal artery or vein laceration or hilar avulsion**; devascularised kidney with active bleeding; shattered kidney

- Advance one grade for multiple injuries, up to grade III.

Treatment

```
Unstable despite resuscitation → laparotomy (± damage control)
                                 or angioembolisation if team ready
Stable → CT staging → non-operative for grade I-V (selected IV-V)
   Active blush → angioembolisation
   Extravasation → observe; stent ± nephrostomy if urinoma
                   enlarges, fever, or pelvis/UPJ injury
```

Non-operative management

- Bed rest, **haemoglobin every 4–6 h**. Succeeds in **>90% of blunt injuries** and selected penetrating ones.
- Failure: instability, or falling haemoglobin despite 2–3 units of blood.
- **Reimage at ~48 h** for grade IV–V with deep lacerations, or with fever, pain or falling haematocrit.
- Urinary extravasation heals spontaneously in **76–90%**.

Angioembolisation (AE)

- Superselective coils for active bleeding, pseudoaneurysm, AVF (arteriovenous fistula) or delayed bleeding.

Indications for exploration

Absolute	Relative
Instability with no or transient response	Extravasation with >25% devitalised parenchyma
Expanding or pulsatile retroperitoneal haematoma	Arterial thrombosis
Suspected pedicle avulsion; UPJ avulsion	Associated colon or pancreatic injury

Operative steps

1. Midline laparotomy. **Control renal vessels before opening Gerota's fascia**: incise the retroperitoneum over the aorta, find the left renal vein crossing it, loop the vein and arteries.
2. **Renorrhaphy**: debride, suture-ligate vessels, close the collecting system watertight, close the defect over a bolster, omental cover, drain. Polar injury: partial nephrectomy.
3. **Nephrectomy** if unreconstructable or unstable. **Damage control**: pack, resuscitate in ICU (intensive care unit), return at 24–48 h.

Renovascular injury

- Unilateral artery thrombosis with a normal other kidney: observe. **Solitary or bilateral: revascularise** (warm ischaemia ideally <120 min) or stent an intimal flap.
- Left renal vein can be ligated (collaterals); the right cannot.

Complications and follow-up

- **Delayed bleeding** (usually within 21 days): AE. Urinoma or abscess: stent, nephrostomy, drain.
- **Hypertension**: Goldblatt (arterial stenosis), **Page kidney** (haematoma compression), AVF.
- **Also know:** AKI (acute kidney injury); hydronephrosis.
- Follow-up: blood pressure, urinalysis, creatinine; EAU: DMSA (dimercaptosuccinic acid) scan for salvaged grade III–V kidneys.

Diagram to draw: left renal vein crossing the aorta, with vessel loops on the vein and artery.

> **Key points to remember:**
>
> - AAST 2018: segmental vessel injury grade IV; main vessel grade V.
> - Image for gross haematuria, microhaematuria with SBP <90, deceleration or penetrating injury.
> - Stable patients, even grade IV–V: non-operative ± angioembolisation.
> - Revascularise only a solitary kidney or bilateral injuries.

Q2. Discuss etiopathogenesis of uric acid calculi. Describe rationale of medical management,

Type: LAQ (25 marks)

Introduction

- Uric acid stones form **8–10%** of stones in the USA and are increasing with obesity and diabetes. They are **radiolucent** and can be **dissolved medically**.

Etiopathogenesis

1. Why humans form uric acid stones

- Humans and apes lack hepatic **uricase**, which converts uric acid to allantoin (10–100 times more soluble). Blood and urine uric acid are therefore high.

2. Physical chemistry: pH is the key

- Uric acid is a weak acid, **pKa about 5.35–5.5**. At pH 5.5 half is free uric acid and half is urate.
- Free uric acid is poorly soluble (about 96–100 mg/L); sodium urate is about 20 times more soluble. At **pH 6.5, over 90% is ionised**.
- **No natural inhibitor** of uric acid crystallisation is known, so stone formation depends almost entirely on supersaturation.

3. The three determinants

Determinant	Causes	Importance
Low urine pH (<5.5)	Insulin resistance, obesity, type 2 diabetes, gout, high animal-protein diet, chronic diarrhoea, ileostomy	**Most important**; most formers have normal uric acid excretion
Low urine volume	Hot climate or occupation, diarrhoea, dehydration	Concentrates uric acid
Hyperuricosuria (>600 mg/day)	Purine-rich diet, gout, myeloproliferative disorders, tumour lysis, uricosuric drugs, inherited defects	Least often the driver in idiopathic cases

4. Why urine is too acid: insulin resistance

- Insulin normally stimulates **ammonia production** from glutamine in the proximal tubule and the Na^+/H^+ exchanger NHE3.
- Insulin resistance reduces ammonium excretion. Acid is then buffered as titratable acid, so urine pH falls.
- **Renal lipotoxicity:** fat accumulates in the proximal tubule, which then burns fatty acids instead of glutamine, so ammoniagenesis falls further.
- Endogenous acid production is also higher. Uric acid stones are thus a **renal manifestation of the metabolic syndrome**; about a third of stones in diabetics are uric acid.

5. Aetiological categories

- **Congenital:** HGPRT (hypoxanthine-guanine phosphoribosyltransferase) deficiency (Lesch–Nyhan), PRPP (phosphoribosylpyrophosphate) synthetase overactivity, hereditary renal hypouricaemia (URAT1, the urate transporter).
- **Idiopathic:** low urine pH without gout ("gouty diathesis").

6. Uric acid and calcium stones

- In **hyperuricosuric calcium oxalate stone disease**, urate promotes calcium oxalate by removing inhibitors, by "salting out" and by **epitaxy** (urate crystals act as a nidus).
- **Also know:** ammonium acid urate stones (laxative abuse, diarrhoea, ileostomy); theobromine (cocoa) inhibits uric acid crystallisation in vitro.

Diagnosis in brief

- Radiolucent on X-ray; CT **200–400 HU (Hounsfield units)**; **dual-energy CT** identifies uric acid.
- Urine pH persistently <5.5; serum uric acid; 24-hour urine; screen for diabetes.

Rationale of medical management

Principle: uric acid stones are a **"pH disease" more than a uric acid disease**. Raising pH turns insoluble uric acid into soluble urate. Diluting urine lowers supersaturation. Lowering uric acid production helps only when excretion is high.

1. Fluids

- Urine output **at least 2.5 L/day** lowers uric acid concentration.

2. Diet and lifestyle

- Limit non-dairy **animal protein** (it gives purines and an acid load; dietary purines provide about 30% of urinary uric acid).
- More fruit and vegetables (alkali load). Weight loss, exercise and glucose control correct the underlying insulin resistance.

3. Urinary alkalinisation (mainstay)

- **Potassium citrate 30–60 mEq/day** in two or three doses. Citrate is metabolised to bicarbonate, raising urine pH.
- **Target pH 6.0–6.5** for prevention (EAU [European Association of Urology]: 6.2–6.8); **avoid pH >7.0**, which risks calcium phosphate stones.
- **Why potassium, not sodium:** sodium alkali raises urinary calcium and can cause calcium stones. Sodium bicarbonate or sodium citrate is used if potassium is unsafe (renal impairment, hyperkalaemia).
- Patients **monitor urine pH with dipsticks**.

4. Dissolution (oral chemolysis)

- Pure uric acid stones dissolve over weeks to months. **EAU: pH 7.0–7.2**; stent or nephrostomy if obstructed or infected; monitor by imaging.

5. Xanthine oxidase inhibitors

- **Allopurinol 300 mg/day** lowers uric acid production. It is **not first-line**, because low pH, not hyperuricosuria, is the usual cause.
- Indications: hyperuricosuria with hyperuricaemia (e.g. gout), recurrence despite alkalinisation (AUA, American Urological Association), hyperuricosuric calcium oxalate stones.
- Reduce the dose in renal impairment. Risks: rash; **allopurinol hypersensitivity syndrome** (2–8% have reactions; may be fatal); Stevens–Johnson syndrome.
- **Febuxostat** is an alternative. **Rasburicase** (recombinant uricase) prevents uric acid nephropathy in tumour lysis.

Landmark trial: Ettinger (1986), randomised placebo-controlled trial: allopurinol more than halved recurrence in hyperuricosuric, normocalciuric calcium oxalate stone formers.

Diagram to draw: sigmoid curve of uric acid solubility against urine pH, marking pKa 5.5 and target 6.0–6.5.

Key points to remember:

- Low urine pH is the main cause; pKa 5.5; no inhibitor exists.
- Insulin resistance reduces ammoniagenesis, so urine is acid.
- Potassium citrate to pH 6.0–6.5 (dissolution 7.0–7.2); avoid >7.0.
- Allopurinol only for hyperuricosuria or failed alkalinisation.

Q3a. Methods for GFR estimation in prospective tx donor.

Type: SAQ (10 marks) (also answered in 2019-Summer Paper 1, Q3a)

Assumed meaning: tx = transplant (prospective living kidney donor).

Why it matters

- The donor loses half the nephron mass; function settles at 60–80% of the pre-donation level. GFR (glomerular filtration rate) is assessed by at least two methods, **including a measured one**.

Methods

1. Serum creatinine and eGFR (estimated GFR)

- Creatinine is freely filtered but depends on **muscle mass, diet and age**, and is also secreted, so it overestimates GFR.
- **CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) 2021** equation: age and sex, **no race term**. MDRD (Modification of Diet in Renal Disease) and Cockcroft–Gault are older.
- Less accurate above 60 mL/min, where donors lie: a screening test only.

2. Cystatin C

- 13-kDa protein from all nucleated cells; freely filtered, then reabsorbed and broken down. Independent of muscle mass; affected by thyroid state, steroids, inflammation.
- The **combined creatinine–cystatin C equation** is more accurate than either alone.

3. 24-hour creatinine clearance

- (Urine creatinine × volume) ÷ serum creatinine; collection errors common.

4. Measured GFR (mGFR): reference

Marker	Method
Inulin	Urinary clearance; gold standard, research only
Iohexol	Plasma clearance; safe and accurate
Iothalamate	Urinary or plasma clearance
^{51}Cr-EDTA (ethylenediaminetetraacetic acid)	Plasma clearance
^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid)	Plasma sampling, or gamma camera (Gates method); purely filtered

5. Split function

- DTPA renogram (or DMSA, dimercaptosuccinic acid) gives each kidney's share. If the difference exceeds 10%, **leave the better kidney with the donor**.

KDIGO (Kidney Disease: Improving Global Outcomes) 2017 decision

GFR (mL/min/1.73 m^2)	Decision
≥90	Accept
60–89	Individualise (age, lifetime risk)
<60	Decline

- Initial eGFR is confirmed by mGFR, measured creatinine clearance or creatinine–cystatin C eGFR.
- Indian practice: DTPA GFR with split function, plus CT (computed tomography) angiography.

Diagram to draw: plasma tracer clearance curve after a single injection.

Key points to remember:

- CKD-EPI 2021 is race-free; cystatin C adds accuracy.
- Inulin is the gold standard; iohexol or DTPA in practice.
- KDIGO 2017: ≥90 accept, 60–89 individualise, <60 decline.

Q3b. Measures to decrease complications of TURP

Type: SAQ (10 marks) (TUR syndrome also answered in 2017-Winter Paper 3, Q3f)

Introduction

- TURP (transurethral resection of the prostate) is the benchmark; overall morbidity approaches one-fifth, mostly preventable.

Preoperative measures

- **Selection:** AUA (American Urological Association) 2023: TURP for glands up to 80 mL; larger glands need HoLEP (holmium laser enucleation) or simple prostatectomy.
- **Sterile urine:** treat UTI (urinary tract infection) first; single-dose prophylaxis covering gram-negatives and enterococci.
- **Anticoagulants:** stop after advice from the prescriber; if they cannot be stopped, choose HoLEP, ThuLEP or PVP (photoselective vaporisation).
- **Finasteride** for recurrent BPH (benign prostatic hyperplasia) bleeding.
- Spinal anaesthesia shows early TUR syndrome; pad the peroneal nerve.

Intraoperative measures

Complication	Prevention
TUR syndrome (hypo-osmolar fluid absorbed, dilutional hyponatraemia)	**Bipolar TURP with saline**; else irrigant ≤60 cm high, resection <90 min, avoid capsular perforation, check sodium in long resections
Bleeding	Systematic resection to capsule; control arterial bleeders as you go; bipolar "cut and seal"; catheter traction for venous ooze
Ureteric orifice injury	Identify orifices first; take a large median lobe down first
Perforation	Respect the capsule; visible fat means perforation
Incontinence	**Stop at the verumontanum**; resect apex last in a bloodless field
Stricture	Calibrate or dilate meatus, lubricate generously, correct sheath size
Bladder neck contracture	Avoid over-resection and excess fulguration; TUIP (transurethral incision of the prostate) for small glands

Postoperative measures

- Large-bore three-way catheter; careful **continuous bladder irrigation** (blocked outflow distends the bladder and worsens bleeding).
- Even routine cases absorb 800–1000 mL of irrigant; check sodium if confused.
- TUR syndrome treatment: **3% saline 100 mL** boluses; raise sodium by no more than 12 mEq/L in 24 h (pontine myelinolysis).
- Stool softeners; no cycling for 4–6 weeks; counsel about **retrograde ejaculation** (about two-thirds).

Diagram to draw: resection field from bladder neck to verumontanum with the external sphincter marked.

> **Key points to remember:**
>
> - Bipolar saline TURP removes TUR syndrome risk; fewer transfusions.
> - Monopolar: irrigant ≤60 cm, resection <90 min, gland <45 mL safer.
> - Verumontanum is the distal limit.

Q3c. What is optical coherence tomography. What is its potential role in management of CA bladder?

Type: SAQ (10 marks) (also answered in Practice Set 5 Paper 4, Q3c and 2017-Summer Paper 4, Q3c)

What is OCT (optical coherence tomography)?

- An **"optical biopsy"** technique: the **optical analogue of B-mode ultrasound**. It measures the echo delay of back-scattered **near-infrared light**, using low-coherence interferometry.
- Gives real-time cross-sectional images with about 10–20 μm resolution but only **1–2 mm depth**. First described in 1991 (ophthalmology).
- **Technique:** a fibre-optic probe passes through the cystoscope working channel and is placed **in contact** with the lesion found by WLC (white-light cystoscopy) or blue light.

Image interpretation

Tissue	OCT appearance
Normal	Three layers: thin low-scatter urothelium, bright lamina propria, darker muscularis
Inflammation	Layers preserved but thickened
Ta papillary tumour	Thick urothelium; lamina propria boundary intact
T1 tumour	**Loss of the urothelium–lamina propria boundary**

Potential role in carcinoma bladder

1. **Diagnosis:** separates benign from malignant flat lesions in real time; fewer unnecessary biopsies.
2. **Staging aid:** suggests Ta versus T1 during TURBT (transurethral resection of bladder tumour).
3. **Resection margins:** checks completeness of TURBT.
4. **Surveillance** of NMIBC (non-muscle-invasive bladder cancer), including after BCG (bacille Calmette–Guérin).
5. **Upper tract:** through the ureteroscope, OCT correlates well with final grade and stage of UTUC (upper tract urothelial carcinoma).

Evidence and limitations

- Small studies: **high sensitivity, lower specificity**. Inflammation, scar and post-BCG change mimic tumour; adding OCT to blue light improves specificity.
- Small field; too shallow for muscle invasion; cannot grade; cost; learning curve. **Investigational.**

Related techniques

- **Blue-light (hexaminolevulinate) cystoscopy:** detects more CIS (carcinoma in situ).
- **NBI (narrow-band imaging):** 415/540 nm light absorbed by haemoglobin.
- **CLE (confocal laser endomicroscopy):** cellular images after fluorescein.

Diagram to draw: OCT image of three bladder layers beside a tumour with loss of layering.

Key points to remember:

- Near-infrared "ultrasound with light"; depth 1–2 mm.
- Loss of layering suggests tumour; boundary loss suggests T1.
- Sensitive but false positives from inflammation; investigational.

Q3d. Tubeless PCNL

Type: SAQ (10 marks) (also answered in 2025-Winter Paper 2, Q3e and 2016-Summer Paper 4, Q3e)

Definition

- PCNL (percutaneous nephrolithotomy) that ends **without a nephrostomy tube**.
- **Tubeless:** drainage by a ureteral stent (double-J, or externalised stent or string). **Totally tubeless:** no stent and no tube.
- **Wickham (1984)** did the first tubeless PCNL; **Bellman (1997)** popularised it.

Rationale

- A large nephrostomy tube causes pain, more analgesia, longer stay and higher cost. Most tracts stop bleeding without tamponade.

Selection

Suitable	Keep a nephrostomy
Complete clearance, single tract	Pyonephrosis or infected stone
Minimal bleeding	Significant bleeding
Intact collecting system	Significant perforation
Supracostal and bilateral cases (shown safe)	Residual stone needing a second look

Technique (exit strategy)

1. Confirm clearance by rigid and **flexible nephroscopy** and fluoroscopy.
2. Antegrade double-J stent, or externalised stent or string.
3. Withdraw the sheath under vision; **safety wire removed last**.
4. Watch the tract; apply pressure. Fibrin glue did not reduce bleeding in randomised trials; some use haemostatic sealants or tract cautery.
5. Urethral catheter overnight.

Evidence

- Meta-analyses of randomised trials: shorter stay and less pain.
- Pain benefit is over large tubes (**>16 Fr**); small tubes (≤10 Fr) give similar comfort without more bleeding.
- No difference in fever, transfusion or stone-free rate.
- Stent discomfort: worse WiSQoL (Wisconsin Stone Quality of Life) scores than nephrostomy in two trials. Externalised stents come out before discharge without cystoscopy.

Disadvantages

- Loss of tract for a second look.
- Stent symptoms; a further procedure for removal.
- Urine leak or urinoma.
- Bleeding without tamponade: reinsert a balloon (Kaye) catheter, then **angioembolisation**.

Current position

- AUA (American Urological Association): choose tubeless, small tube or stent case by case. Pairs well with mini-PCNL.

Diagram to draw: kidney with sealed tract and double-J stent.

Key points to remember:

- Tubeless = stent only; totally tubeless = neither.
- Benefit is mainly over large (>16 Fr) tubes.
- Keep a tube for bleeding, perforation, infection or second look.

Q3e. Medullary sponge kidney

Type: SAQ (10 marks)

Definition

- **MSK (medullary sponge kidney; precalyceal canalicular ectasia):** dilated (ectatic) distal **collecting ducts** with small cysts and diverticula **confined to the medullary pyramids**. Nephrons are initially normal.
- Usually **non-heritable** (sporadic); usually bilateral.

Pathogenesis of stones and nephrocalcinosis

- **Stasis and infection** in the ectatic ducts favour crystal deposition.
- **Tubular defects:**
  - **Hypercalciuria** in one-third to one-half (mostly absorptive; sometimes renal leak).
  - **Incomplete distal RTA (renal tubular acidosis)** in 30–40%, shown on acid loading. This causes **hypocitraturia** and alkaline urine.
  - Impaired urinary concentration.
  - Mutations in H^+-ATPase genes (ATP6V1B1, ATP6V0A4) support an MSK–distal RTA link.
- Result: calcium stones within dilated ducts and medullary **nephrocalcinosis**.

Clinical features

- Often silent for life; presents **after 20 years**.
- **Renal colic 50–60%**, UTI (urinary tract infection) 20–33%, gross haematuria 10–18%.
- Found in **2.6–21% of stone formers**, more often women.

Diagnosis

- **Urographic triad:**
  1. Enlarged kidneys, with or without papillary calcification.
  2. Elongated papillary tubules or cavities filling with contrast: **"bristles on a brush"** (paintbrush); when calcified, a **"bouquet of flowers"**.
  3. Papillary blush and persistent medullary opacification.
- CT urography now replaces IVU (intravenous urography); plain CT shows medullary nephrocalcinosis.
- **Metabolic work-up:** 24-hour urine (calcium, citrate, pH), serum bicarbonate and potassium.
- **Differentials:** papillary necrosis, other medullary nephrocalcinosis (distal RTA, hyperparathyroidism), calyceal diverticulum.

Management

Agent	Target	Note
Potassium citrate	Hypocitraturia, hypercalciuria	**First choice**; reduces calciuria and recurrence
Thiazide	Hypercalciuria	Useful even without hypercalciuria
Inorganic phosphate	Hypercalciuria	If thiazide unsuitable; avoid with urease infection
Fluids, low-sodium diet	Volume, calciuria	All patients

- Treat infection; stones by SWL (shock-wave lithotripsy) or ureteroscopy; stones inside ducts may be unreachable.
- **Prognosis:** usually benign; renal failure is rare.

Diagram to draw: pyramid with ectatic ducts filled with contrast ("brush") and small calculi.

Key points to remember:

- Ectatic collecting ducts; sporadic; "bristles on a brush".
- Hypercalciuria, incomplete distal RTA, hypocitraturia.
- Potassium citrate is first-line prevention.

Q3f. Amendments to THOA: 2011.

Type: SAQ (10 marks)

Indian law; not covered in the chapter notes.

Background

- **THOA (Transplantation of Human Organs Act) 1994** regulated organ transplantation, banned commercial dealing and legalised **brain-stem death**.
- Organ trade continued and deceased donation stayed low. The **Transplantation of Human Organs (Amendment) Act 2011** responded; **Rules followed in 2014**.

Main amendments

Area	Change
Scope	**Tissues** added; renamed the **Transplantation of Human Organs and Tissues Act (THOTA)**
Near relatives	**Grandparents and grandchildren** added to spouse, children, parents and siblings
Swap donation	**Paired exchange** between near-relative pairs allowed, with Authorisation Committee approval
Retrieval centres	Hospitals with ICU (intensive care unit) facilities can register as **non-transplant organ retrieval centres**; **tissue banks** registered
Transplant coordinator	Mandatory in every registered transplant hospital
Mandatory ("required") request	ICU doctor must ask relatives of a potential brain-dead donor about donation and inform the retrieval centre
Brain-death board	If a neurologist or neurosurgeon is unavailable, a surgeon, physician, anaesthetist or intensivist nominated by the hospital can serve
Eye donation	Trained technicians may remove corneas
Registry and network	National human organ and tissue removal and storage network and **national registry** of donors and recipients (now run by NOTTO, the National Organ and Tissue Transplant Organisation)
Penalties	Raised (see below)

Penalties under the amendment

- **Removal of organs without authority:** imprisonment up to **10 years** and fine up to ₹20 lakh.
- **Commercial dealing in organs:** imprisonment **5–10 years** and fine **₹20 lakh to ₹1 crore**.
- Registered medical practitioner: name removed from the register for **3 years** for a first offence and **permanently** for a later offence.

Safeguards retained

- Unrelated donors and foreigners need **Authorisation Committee** approval.
- Minors and mentally ill persons cannot be living donors except in exceptional, approved cases.

Diagram to draw: flowchart: potential donor in ICU → required request → transplant coordinator → brain-stem death board → retrieval → national registry and allocation.

> **Key points to remember:**
>
> - 2011: tissues added; grandparents and grandchildren became near relatives.
> - Swap donation, retrieval centres, transplant coordinators, required request.
> - Commercial dealing: 5–10 years and ₹20 lakh to ₹1 crore.

Rapid Revision Sheet – Practice Set 9 Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Diagnosis and Treatment of renal trauma (LAQ)

- AAST 2018: segmental vessel injury grade IV; main vessel grade V.
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- Allopurinol only for hyperuricosuria or failed alkalinisation.

Q3a. Methods for GFR estimation in prospective tx donor. (SAQ)

- CKD-EPI 2021 is race-free; cystatin C adds accuracy.
- Inulin is the gold standard; iohexol or DTPA in practice.
- KDIGO 2017: ≥90 accept, 60–89 individualise, <60 decline.

Q3b. Measures to decrease complications of TURP (SAQ)

- Bipolar saline TURP removes TUR syndrome risk; fewer transfusions.
- Monopolar: irrigant ≤60 cm, resection <90 min, gland <45 mL safer.
- Verumontanum is the distal limit.

Q3c. What is optical coherence tomography. What is its potential role in management of CA bladder? (SAQ)

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